

Total Error: Using ML to Measure Total Exposure

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Abstract

Accurate domain classification need not yield accurate measures of how much content people consume. Each classification error is multiplied by the user’s visits or time on that domain. Unequal browsing volumes can therefore turn small systematic errors into large, unequal biases in exposure totals. Because users share domain labels, their measurement errors are also correlated. I derive the bias and covariance of exposure totals and show how both enter group comparisons and regressions. A simulation isolates the accumulation of bias; the domain-level formulation explains why uncertainty must also follow the shared labels. Validation and manual labeling should reflect the exposure measure and comparison the researcher wants to estimate.

Say we want to measure how much pornography people consume online. A model labels domains, and browsing records supply visits or time spent on each domain. Summing visits to domains labeled pornographic gives a measure of each person’s exposure. But a classifier’s error rate gives equal weight to domains, whereas an exposure total weights each domain by use. A wrong label on a heavily visited domain can distort many people’s totals at once.

This is related to quantification, which estimates label prevalence in a target population from imperfect predictions (Vaz et al., 2019). Here the objects of interest are person-specific weighted totals and comparisons among them. Different users attach different weights to the same predictions. That structure determines both bias and dependence across users; correcting an overall prevalence does not by itself validate a user-level comparison.

1 From domain labels to exposure

Let $i = 1, \dots, n$ index users and $j = 1, \dots, k$ index domains. Let $c_{ij} \geq 0$ be visits by user i to domain j , $y_j \in \{0, 1\}$ the true domain label, and $\hat{y}_j \in \{0, 1\}$ the predicted label. Define

$$T_i = \sum_j c_{ij} y_j, \quad \hat{T}_i = \sum_j c_{ij} \hat{y}_j, \quad V_i = \sum_j c_{ij}, \quad D_i = \sum_j \mathbf{1}\{c_{ij} > 0\}. \quad (1)$$

T_i counts visits to positive domains; V_i counts all visits; D_i counts distinct visited domains. Replacing visits by duration changes the unit to time and leaves the algebra unchanged. These quantities measure exposure to domains with a given label. They do not establish which pages or content a person consumes on a domain that hosts several kinds of content.

Write $e_j = \hat{y}_j - y_j$, let C be the matrix of browsing weights, and let $d = \hat{T} - T$. The exact measurement-error identity is

$$d = Ce, \quad d_i = \sum_j c_{ij} e_j. \quad (2)$$

*gsood07@gmail.com. The argument about browsing volume and costly errors on popular domains appears in the June 2020 manuscript coauthored with Suriyan Laohaprapanon (Laohaprapanon and Sood, 2020). The standalone note and original simulation were added in June 2023. This revision corrects the error-rate definitions and develops the covariance and inference results. Code and history: https://github.com/soodoku/total_error.

The same e_j enters every user who visits domain j . Repeated visits multiply one error; they are not independent opportunities to classify the domain again.

For expectations and variances, fix C and y and posit repeated prediction errors, such as those induced by repeated training samples or a specified error model. Let $b = E[e \mid C, y]$ and $\Sigma = \text{Cov}(e \mid C, y)$. Then

$$E[d \mid C, y] = Cb, \quad \text{Cov}(d \mid C, y) = C\Sigma C^\top. \quad (3)$$

For a fixed fitted classifier and fixed population, d is fixed. A standard error requires a stated source of randomness. Equations (3) describe uncertainty from the specified prediction-error process; they do not include user sampling, browsing-log error, or uncertainty about the construct.

2 Bias and the denominators of error rates

Let α_j be the probability of predicting one for a truly negative domain and β_j the probability of predicting zero for a truly positive domain, conditional on the browsing design. Then

$$b_j = \alpha_j(1 - y_j) - \beta_j y_j, \quad E[d_i \mid C, y] = \sum_j c_{ij} \{\alpha_j(1 - y_j) - \beta_j y_j\}. \quad (4)$$

False-positive rates condition on true negatives; false-negative rates condition on true positives. Multiplying a false-positive rate by predicted positives uses the wrong denominator. Such a calculation would instead require the probability of a true negative conditional on a positive prediction, together with assumptions that justify transferring that probability to the weighted population. These conditional probabilities are not interchangeable.

If $\alpha_j = \alpha$ on negative domains and $\beta_j = \beta$ on positive domains,

$$E[d_i \mid C, y] = \alpha(V_i - T_i) - \beta T_i = V_i \{\alpha - (\alpha + \beta)\pi_i\}, \quad \pi_i = T_i/V_i, \quad (5)$$

where the share π_i is defined for $V_i > 0$. At a fixed true share, expected bias scales with browsing volume unless the expression in braces is zero. For $\alpha + \beta > 0$, cancellation occurs at $\pi_i = \alpha/(\alpha + \beta)$. Equal positive error rates cancel only when half of visits are to truly positive domains. Thus the difference between FPR and FNR alone does not determine bias. Nor does D_i alone: repeated visits matter through c_{ij} .

The common-rate assumption is stronger than knowing an average error rate on a validation set of domains. Errors may depend on popularity, domain type, or features associated with a particular group's browsing. In that case, equation (4), with rates conditional on those features, is the relevant expression. The bias in a share is $E[d_i \mid C, y]/V_i$; dividing by volume removes the mechanical scale factor, not misclassification bias.

3 Shared errors and standard errors

If prediction errors are independent across domains conditional on C, y , let $v_j = \text{Var}(e_j \mid C, y)$. Equation (3) becomes

$$\text{Var}(d_i \mid C, y) = \sum_j c_{ij}^2 v_j, \quad \text{Cov}(d_i, d_\ell \mid C, y) = \sum_j c_{ij} c_{\ell j} v_j. \quad (6)$$

For binary predictions, $v_j = \alpha_j(1 - \alpha_j)$ on negative domains and $v_j = \beta_j(1 - \beta_j)$ on positive domains. Variance depends on squared visits. Treating repeated visits as independent would replace

these squared weights by visit counts and miss the repeated use of the same label. Different users have correlated errors whenever they share uncertain domains. Errors across domains may also be correlated, for example through a shared training sample; then the full Σ in equation (3) is needed.

Let a comparison be $\theta = a^\top T$ for fixed user weights a . For a difference of group means, $a_i = 1/n_A$ in group A and $a_i = -1/n_B$ in group B . Set $w = C^\top a$, the domain weights induced by the comparison. Then

$$\hat{\theta} - \theta = w^\top e, \quad \text{Bias}(\hat{\theta} \mid C, y) = w^\top b, \quad \text{Var}(\hat{\theta} - \theta \mid C, y) = w^\top \Sigma w. \quad (7)$$

A popular domain can contribute large errors to individual totals yet cancel from a group contrast when the groups have identical average usage of it. Conversely, an uncertain domain visited mostly by one group can dominate the comparison. Bias and variance must therefore be assessed for the intended contrast, not inferred from overall domain accuracy. Dropping covariance terms can raise or lower a contrast’s variance: shared errors may reinforce within groups and cancel between them.

For a full-rank, fixed regressor matrix X , consider the least-squares coefficient that would be obtained using the true exposure outcome T . Writing $A = (X^\top X)^{-1} X^\top$, prediction changes that coefficient by ACe . Its conditional bias is ACb and its additional measurement-error covariance is

$$AC\Sigma C^\top A^\top. \quad (8)$$

Mean-zero measurement error can therefore affect standard errors even when it does not bias the coefficient. Systematic error associated with regressors can affect both. This statement concerns exposure as an outcome; using noisy exposure as a regressor is a different errors-in-variables problem (Fong and Tyler, 2021).

These expressions quantify the prediction-error component of uncertainty. They cannot simply be added to an arbitrary reported regression covariance: the components may overlap or be dependent. A combined analysis must specify sampling and prediction jointly. Resampling users while holding predicted labels fixed omits repeated-training uncertainty. To propagate domain-label uncertainty, each replicate must use one common label draw per domain for all users, retaining cross-domain dependence where required. Neither larger user samples nor user-level clustering alone guarantees that this component vanishes. Standard errors also do not repair systematic bias.

4 A connection to label sampling

The same mapping supplies a remedy when a researcher can obtain verified labels for a probability sample of domains. The companion `fewlab` project develops sampling designs for reusable labels (Sood, 2026). Its regression decomposition is $H = AC$, with column h_j giving the influence of domain j on the coefficient. For exposure shares, replace C by its row-normalized version. For the scalar contrast in equation (7), the corresponding influence is w_j .

Freeze predictions q_j before drawing a label sample. Let R_j indicate selection, with known inclusion probability $s_j > 0$ for every domain with nonzero influence. Conditional on the fixed population and predictions, the difference estimator is

$$\tilde{\theta} = \sum_j w_j q_j + \sum_j \frac{R_j}{s_j} w_j (y_j - q_j). \quad (9)$$

Since $E_R[R_j/s_j] = 1$, its expectation over label sampling is $\sum_j w_j y_j = \theta$, regardless of the classifier’s errors. Verified labels must measure y_j correctly. Retraining the proxy on the same sampled labels requires a separate argument; the fixed-proxy proof does not cover it.

Under independent Bernoulli domain sampling,

$$\text{Var}_R(\tilde{\theta}) = \sum_j \frac{1 - s_j}{s_j} w_j^2 (y_j - q_j)^2. \quad (10)$$

The realized sample gives an unbiased variance estimator by replacing each summand with $R_j(1 - s_j)w_j^2(y_j - q_j)^2/s_j^2$. Fixed-size or balanced designs require their joint-inclusion terms or a justified approximation. An inaccurate proxy can increase variance relative to omitting it, but does not introduce design bias under these assumptions. This is uncertainty from randomized label acquisition, with y and q fixed; it is distinct from the prediction-error process in equation (3). Both calculations preserve the reuse of a domain label across people.

5 A simulation of accumulating bias

The original simulation draws 1,000 respondents, each with an independent observation count uniformly distributed over the integers from 5 to 1,000. Within respondents, independent bivariate normal pairs have zero means, unit variances, and correlation $\rho = 0.9$. The first component is truth and the second a proxy. The original code adds 0.1 times the respondent’s sample standard deviation of truth to every proxy observation. Table 1 retains those random draws and compares that specification with no added bias and an exactly constant addition of $b = 0.1$.

Table 1: Correlations with truth and signed error in respondent totals

Proxy	Correlation			Mean total error
	Observation	Mean	Total	
No added bias	0.900	0.903	0.893	0.139
Constant bias	0.900	0.903	0.537	51.236
Original SD-scaled bias	0.900	0.901	0.538	51.239

Each row uses the same 1,000 respondents and random draws (seed 1234). Error is proxy minus truth. Measurements are centered continuous variables, so totals and errors are in arbitrary units. These are descriptive summaries of one simulated dataset; the table does not estimate uncertainty in browsing data.

In the original specification, the observation-level correlation is 0.900 and the total correlation is 0.538. Mean total error is 51.239. Adding an exactly constant bias leaves the correlations of observations and respondent means unchanged. The SD-scaled addition only approximates a constant, so that invariance is not exact for the original specification.

The constant-bias design has a simple population expression. Let N be the observation count, independent of the pairs (Y_r, Z_r) , and set $S = \sum_{r=1}^N Y_r$, $\hat{S} = \sum_{r=1}^N (Z_r + b)$. Because the pairs have zero means and unit variances,

$$\begin{aligned} \text{Var}(S) &= \text{E}[N], & \text{Var}(\hat{S}) &= \text{E}[N] + b^2 \text{Var}(N), & \text{Cov}(S, \hat{S}) &= \rho \text{E}[N], \\ \text{Corr}(S, \hat{S}) &= \frac{\rho}{\sqrt{1 + b^2 \text{Var}(N)/\text{E}[N]}}. \end{aligned} \quad (11)$$

The population correlation is 0.553, compared with 0.537 in the constant-bias simulation. With a fixed observation count, bias shifts every total equally and does not change the correlation. For nonzero means or a relation between counts and measurements, equation (11) need not hold. Accumulating bias does not universally reduce correlations.

This simulation isolates bias accumulation. Its normal draws can be negative, so they are not literal visit counts or durations. It also generates observations independently across respondents

and does not simulate shared domain errors. The covariance results above follow from the domain formulation, and are checked separately by enumerating all prediction vectors in a small binary example.

6 Correction and validation

Under the common-rate model with known α, β and $\alpha + \beta \neq 1$, equation (5) gives an unbiased correction:

$$\tilde{T}_i = \frac{\hat{T}_i - \alpha V_i}{1 - \alpha - \beta}, \quad \mathbb{E}[\tilde{T}_i \mid C, y] = T_i. \quad (12)$$

The correction is unstable when the denominator is close to zero and may return values outside $[0, V_i]$. Clipping changes its expectation. Estimating error rates introduces additional uncertainty shared across users; rates estimated on unweighted domains need not transfer to weighted exposure. This is the connection to classifier-based quantification (Vaz et al., 2019), with the additional requirement that the rates apply to each user’s weights.

Using probabilities q_j instead of hard labels yields $\sum_j c_{ij} q_j$. Under a conditional-label model with information $\mathcal{I}$ that includes browsing weights, $q_j = \Pr(y_j = 1 \mid \mathcal{I})$ makes that sum $\mathbb{E}[T_i \mid \mathcal{I}]$. Marginal calibration across domains alone does not supply this conditional property for every user or group. This conditional-label interpretation is distinct from the repeated-prediction interpretation used above; an uncertainty calculation must choose and state its target.

Validation should match the research quantity. For individual totals, large c_{ij} magnify label errors. For a particular contrast under independent domain errors, the variance contribution is $w_j^2 v_j$ and the bias contribution is $w_j b_j$. Those quantities can guide a labeling audit more directly than domain frequency alone. Verified labels for influential domains eliminate their classification error if those labels are correct. Usage-weighted training is another option, but its effect on exposure error requires validation; concentrated weights do not by themselves establish that model performance will worsen.

Changing classification thresholds is a sensitivity analysis, not a set of coverage-guaranteed bounds on true exposure. Winsorizing visits reduces the influence of large weights but changes the exposure quantity being estimated. The most useful validation reports errors in totals and in the intended comparisons, together with uncertainty that preserves the common domain labels.

References

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